

Six-Cycle Insertion and Hook-Unimodality in Higher Lie Characters

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Abstract

Let $M_\lambda(x)$ denote the hook-multiplicity polynomial of the higher Lie character attached to a conjugacy class of cycle type λ . Adin–Hegedűs–Roichman conjectured that the coefficient sequence of $M_\lambda(x)$ is unimodal for every partition λ . Using their product formula for rectangular cycle types, we prove the conjecture for the infinite family (6^s) for every $s \geq 1$. In fact, $(1+x)M_{(6^s)}(x)$ is strictly log-concave on its nonzero support. We then prove a six-cycle insertion theorem: adjoining any positive number of six-cycles preserves unimodality when the remaining cycle type is unimodal and contains no six. As a consequence, every nonempty partition all of whose parts lie in $\{1, 2, 3, 4, 5, 6\}$ has a unimodal hook-multiplicity sequence, with no bound on multiplicities or total size.

1 Background and statement of the results

For a partition $\lambda \vdash n$, write

$$M_\lambda(x) = \sum_{k=0}^{n-1} m_{(n-k, 1^k), \lambda} x^k,$$

where $m_{(n-k, 1^k), \lambda}$ is the multiplicity of the hook character $\chi^{(n-k, 1^k)}$ in the higher Lie character ψ^λ . Adin–Hegedűs–Roichman [1] conjectured that the coefficient sequence of $M_\lambda(x)$ is unimodal for every λ (their Conjecture 8.1). They verified the conjecture for all $|\lambda| \leq 15$ and for rectangular shapes (r^s) with $r \leq 40$ and $s \leq 5$.

We use two identities from [1]. First, when nonempty partitions λ and ν have disjoint sets of part sizes,

$$M_{\lambda \sqcup \nu}(x) = (1+x)M_\lambda(x)M_\nu(x). \quad (1)$$

Second, for fixed r ,

$$1 + (1+x) \sum_{s \geq 1} M_{(r^s)}(x) y^s = \prod_{j=0}^r (1 - (-x)^j y)^{(-1)^{j+1} f_j(r)}, \quad (2)$$

where

$$f_j(r) = \frac{1}{r} \sum_{d \mid \gcd(r, j)} \mu(d) (-1)^{(d+1)j/d} \binom{r/d}{j/d}.$$

For $r = 6$,

$$(f_0(6), \dots, f_6(6)) = (0, 1, 3, 3, 2, 1, 0),$$

so

$$E_6(x, y) := 1 + (1+x) \sum_{s \geq 1} M_{(6^s)}(x) y^s = \frac{(1+xy)(1+x^3y)^3(1+x^5y)}{(1-x^2y)^3(1-x^4y)^2}. \quad (3)$$

Theorem 1 (all six-cycles). *For every integer $s \geq 1$, $M_{(6^s)}(x)$ has unimodal coefficients. Moreover, $(1+x)M_{(6^s)}(x)$ has no internal zeros and is strictly log-concave between its first and last nonzero terms.*

Theorem 2 (six-cycle insertion). *Let ν be a nonempty partition with no part equal to 6. If $M_\nu(x)$ is unimodal, then for every $s \geq 1$, $M_{\nu \sqcup (6^s)}(x)$ is unimodal.*

Corollary 1. *Every nonempty partition whose parts belong to $\{1, 2, 3, 4, 5, 6\}$, at arbitrary multiplicities and arbitrary total size, has a unimodal hook-multiplicity sequence.*

2 An exact coefficient formula for (6^s)

Write

$$M_{(6^s)}(x) = x^{2s-1}P_s(x), \quad P_s(x) = \sum_{k=0}^{2s+1} q_{s,k}x^k.$$

The coefficients are

$$q_{s,0} = \frac{s(s+1)}{2}, \tag{4}$$

$$q_{s,1} = \frac{3s^2 - s + 2}{2}, \tag{5}$$

$$q_{s,2} = \frac{5s^2 - 7s + 4}{2}, \tag{6}$$

and for $3 \leq k \leq 2s$,

$$q_{s,k} = Q_s(k) := \frac{k^3 - 4k^2s - 3k^2 + 4ks^2 + 8ks + 6k - 4s^2 - 6s - 4}{2}, \tag{7}$$

while

$$q_{s,2s+1} = s. \tag{8}$$

For completeness, an exact rational identity yielding these formulas is

$$P_s(x) = \frac{N_0(s, x) + x^{2s}N_1(s, x)}{(1-x)^4}, \tag{9}$$

where

$$\begin{aligned} a_2(x) &= \frac{1}{2}(1 - x - x^2 + 2x^3 - x^4 - x^5 + x^6), \\ a_1(x) &= \frac{1}{2}(1 - 5x + 3x^2 - 3x^4 + 5x^5 - x^6), \\ a_0(x) &= x - 2x^2 + 5x^3 - 2x^4 + x^5, \\ b_1(x) &= x^5 - x^4 + x^2 - x, \\ b_0(x) &= -2x^4 + x^3 - 2x^2, \end{aligned}$$

with $N_0(s, x) = a_2(x)s^2 + a_1(x)s + a_0(x)$ and $N_1(s, x) = b_1(x)s + b_0(x)$. Summing (9) over $s \geq 1$ after setting $z = x^2y$ and using the standard sums for $\sum z^s$, $\sum sz^s$, and $\sum s^2z^s$ reproduces (3). Coefficient extraction from $(1-x)^{-4}$ gives the displayed $q_{s,k}$ formulas.

3 Unimodality of the six-cycle family

For $s = 1, 2$ the nonzero coefficient sequences of P_s are respectively

$$(1, 2, 1, 1), \quad (3, 6, 5, 5, 4, 2),$$

which are unimodal.

Assume $s \geq 3$. The first three forward differences are

$$s^2 - s + 1, \quad s^2 - 3s + 1, \quad \frac{(3s - 5)(s - 2)}{2},$$

all positive. On the cubic interval, using (7),

$$D_s(k) := Q_s(k + 1) - Q_s(k) = \frac{3k^2 - 8ks - 3k + 4s^2 + 4s + 4}{2} \quad (10)$$

for $3 \leq k \leq 2s - 1$. This is a convex quadratic in k , and

$$D_s(2s - 1) = 5 - 3s < 0.$$

Therefore its nonpositive sublevel set on the interval is contiguous and contains the final point: once a nonpositive forward difference appears, no later forward difference can be positive. Finally,

$$q_{s,2s+1} - q_{s,2s} = 2 - 2s < 0.$$

Hence the coefficient sequence increases and then decreases, with a possible plateau. Thus P_s , and hence $M_{(6^s)}$, is unimodal.

4 Strict log-concavity after one factor $(1 + x)$

Let

$$B_s(x) = (1 + x)P_s(x) = \sum_{k=0}^{2s+2} b_{s,k} x^k,$$

and put

$$L_{s,k} = b_{s,k}^2 - b_{s,k-1}b_{s,k+1}.$$

For $s = 1$, B_1 has coefficients $(1, 3, 3, 2, 1)$ and the internal $L_{s,k}$ are $(6, 3, 1)$. For $s = 2$, the coefficients are $(3, 9, 11, 10, 9, 6, 2)$ and the corresponding differences are $(48, 31, 1, 21, 18)$.

Now let $s \geq 3$. The first coefficients are

$$b_0 = \frac{s(s+1)}{2}, \quad b_1 = 2s^2 + 1, \quad b_2 = 4s^2 - 4s + 3, \quad b_3 = \frac{13s^2 - 25s + 18}{2}.$$

For $4 \leq k \leq 2s$,

$$b_k = R_s(k) := \frac{2k^3 - 8k^2s - 9k^2 + 8ks^2 + 24ks + 21k - 12s^2 - 24s - 18}{2}. \quad (11)$$

The last two coefficients are $b_{2s+1} = 4s - 2$ and $b_{2s+2} = s$.

At the first four positions set $u = s - 3 \geq 0$. Direct expansion gives

$$\begin{aligned} L_{s,1} &= 2u^4 + 24u^3 + \frac{225}{2}u^2 + \frac{483}{2}u + 199, \\ L_{s,2} &= 3u^4 + 29u^3 + \frac{229}{2}u^2 + \frac{433}{2}u + 159, \\ L_{s,3} &= \frac{9}{4}u^4 + \frac{33}{2}u^3 + \frac{233}{4}u^2 + 106u + 63, \\ L_{s,4} &= 9u^4 + 61u^3 + 207u^2 + 335u + 181. \end{aligned}$$

All are positive. For $5 \leq k \leq 2s - 1$, set $u = k - 5 \geq 0$ and $v = 2s - k - 1 \geq 0$. Expansion of (11) gives

$$\begin{aligned} L_{s,k} &= 2u^2v^2 + 10u^2v + 7u^2 + 14uv^2 + 67uv + \frac{83}{2}u \\ &\quad + v^4 + 10v^3 + \frac{123}{2}v^2 + 172v + 118 > 0. \end{aligned}$$

At the final two positions, again with $u = s - 3$,

$$L_{s,2s} = 13u^2 + 50u + 64 > 0, \quad L_{s,2s+1} = 7u^2 + 35u + 46 > 0.$$

Thus B_s is strictly log-concave on its nonzero support. Its endpoint coefficients are positive, so it has no internal zeros.

5 A convolution lemma and six-cycle insertion

Lemma 1. *Let c be a finite nonnegative log-concave sequence with no internal zeros and let a be a finite nonnegative unimodal sequence. Then the convolution $c * a$ is unimodal.*

Proof. Extend both sequences by zero to all integers. Log-concavity and contiguous support imply the total-positivity inequality

$$c_{\ell-i}c_{k-j} \leq c_{k-i}c_{\ell-j} \quad (i < j, \ k < \ell).$$

Indeed, when the left side is positive this follows by multiplying the nonincreasing successive ratios of c ; when it is zero the inequality is immediate.

Let $d_j = a_j - a_{j-1}$. Since a is unimodal, all positive terms of d precede all negative terms. Put

$$P_k = \sum_i d_i^+ c_{k-i}, \quad N_k = \sum_j d_j^- c_{k-j}.$$

The forward difference of $c * a$ at position k is $P_k - N_k$. Multiplying the displayed total-positivity inequality by $d_i^+ d_j^-$ and summing gives

$$P_\ell N_k \leq P_k N_\ell \quad (k < \ell).$$

A negative forward difference at k followed by a positive forward difference at ℓ would require $N_k > P_k$ and $P_\ell > N_\ell$, giving the opposite strict product inequality. Hence the forward differences change sign at most once, and $c * a$ is unimodal. $\square$

Now let ν be nonempty, unimodal, and contain no part 6. By (1),

$$M_{\nu \sqcup (6^s)}(x) = ((1+x)M_{(6^s)}(x))M_\nu(x).$$

The first factor has a strictly log-concave coefficient sequence with no internal zeros, while the second is nonnegative and unimodal. The lemma proves the six-cycle insertion theorem.

6 Partitions with parts at most six

For positive multiplicities, (2) gives

$$M_{(1^a)} = 1, \quad M_{(2^b)} = x^{2b-1}, \quad M_{(3^c)} = x^{2c-1},$$

$$M_{(4^d)} = d x^{2d-1} (1 + x),$$

and

$$M_{(5^t)} = x^{2t-1} \left[t + \sum_{j=1}^{2t-1} (2t-j)x^j + x^{2t} \right]. \quad (12)$$

The nonzero coefficient sequence in the bracket of (12) is

$$t, \ 2t-1, \ 2t-2, \dots, 1, 1,$$

which is unimodal. Repeated use of (1) combines distinct part sizes through multiplication by $1+x$, whose coefficient sequence is log-concave. The convolution lemma therefore proves unimodality for every partition with parts at most five. If the partition also contains sixes, remove all six-parts. If a nonempty remainder remains, apply the six-cycle insertion theorem; if no remainder remains, apply the all-six theorem. This proves the corollary.

7 Scope

The results above prove an infinite subfamily of Conjecture 8.1 of [1]; they do not prove the conjecture for arbitrary partitions. They also do not claim historical priority beyond what can be established by a complete literature review.

References

- [1] R. M. Adin, P. Hegedűs, and Y. Roichman, *Higher Lie characters and cyclic descent extension on conjugacy classes*, Algebraic Combinatorics **6** (2023), no. 6, 1557–1591. DOI: 10.5802/alco.323.